

11. Individual Contribution Statement & Team Responsibilities

11.1 Project Team Identity

1. **Project Name:** Project 1 – ATX Bench Power Supply
2. **Course / Laboratory:** Electronic Class Project Laboratory
3. **Source Hardware:** PLENTY COMPUTER (Model: ATX500WS, Max Output Power: 500W)
4. **Repository Revision:** Rev 1.0 (As-Built Initial Submission)

Team Members:

1. **Puttimeth Jaovajavanil** (Student ID: 6809107660041) —
Role: Role: Lead Documentation & Quality Assurance
2. **Kongphop Konghirun** (Student ID: 6809107660083) —
Role: Lead Systems Hardware & Safety

11.2 Task Distribution & Detailed Responsibilities

Kongphop Konghirun (Student ID: 6809107660083)

- **Primary Responsibilities:**
 - **Hardware Integration & Assembly:** Executed physical component wiring, breakout board installation, banana jack mounting, and internal wiring harness routing inside the acrylic project enclosure.
 - **Safety Boundary Enforcement:** Implemented branch fuse holders, crimped insulated heat-shrink sleeves on all rear terminals, and verified cable strain relief grommets.
 - **Electrical Calculations:** Performed wire gauge (18 AWG) voltage drop calculations, branch fuse sizing schedules, and buck-boost converter thermal/loss estimations.

- **Unpowered Safety Testing:** Led the Pre-Power Verification Checklist (Doc 6) using digital multimeter resistance and continuity checks.
- **Estimated Contribution Ratio: 50%**

Puttimeth Jaovajavanil (Student ID: 6809107660041)

- **Primary Responsibilities:**
 - **Repository & Workflow Management:** Configured GitHub repository structure, Version Control workflows, relative link trees in [README.md](#), and PDF release pipelines.
 - **Schematic & Layout Drafting:** Drafted the as-built low-voltage distribution schematic diagram (Doc 4) and front-panel enclosure dimension drawings.
 - **BOM & Component Verification:** Documented component datasheets, part numbers, cost schedules, and rating verifications for the Bill of Materials (Doc 5).
 - **Verification & Test Evidence:** Recorded functional testing logs, diagnostic fault evaluations, and operational guidelines (Docs 8, 9, 10).
- **Estimated Contribution Ratio: 50%**

11.3 Collaborative Working Summary & Joint Verification

Both team members contributed equally to the physical build, safety audits, laboratory testing, and documentation repository. Key milestones achieved through active joint collaboration include:

1. **Safety Boundary Compliance:** Both members inspected the unmodified PLENTY COMPUTER PSU metal casing to enforce closed-casing safety limits.

2. **Testing & Retesting:** Joint execution of unpowered insulation checks, load regulation testing, ripple monitoring, and simulated fault diagnostic retests.
3. **Documentation Peer Review:** Cross-review of technical text, markdown structures, and PDF exports prior to Git commits.